5

Carbohydrates

INTRODUCTION

Carbohydrates are the primary source of energy in diets fed to dairy cattle and usually comprise 60 to 70 percent of the diet. Besides being the primary energy source to the ani- mal, carbohydrates provide substrate for growth of ruminal microbes and production of microbial protein, and fibrous carbohydrates help maintain proper gastrointestinal func- tion. The carbohydrate fraction of feeds is a diverse mixture ranging from polymers to simple sugars that are usually partitioned according to their digestion characteristics in the animal. Carbohydrates are broadly classified by their solubility in neutral detergent, including the insoluble neu- tral detergent fiber (NDF) fraction and the neutral detergent soluble fraction (see Figure 5-1). The NDF carbohydrates are further categorized as forage NDF (fNDF) or nonforage NDF (nfNDF), each containing hemicellulose, cellulose, and lignin. The neutral detergent-soluble carbohydrates (NDSC) are divided into starch, water-soluble carbohydrates (WSC; e.g., fructans, sugars), and neutral detergent-soluble fiber (NDSF; e.g., pectic substances).

Availability to the animal varies by carbohydrate fraction and source of the carbohydrate (e.g., type of feed, processing, growing environment). Cattle have the capacity to digest starch, lactose, microbial glycogen, and trehalose (disaccha- ride) based on enzymes present in the pancreatic secretions and small intestinal mucosa (Kreikemeier et al., 1990). All other carbohydrates, including sucrose and those in NDF, must be degraded and utilized by gastrointestinal microbes for them to provide nutrients to the animal. Total-tract di- gestibility of the carbohydrate fractions affects total nutrient supply to the animal, whereas ruminal fermentation affects the type and temporal supply of fuels, which affect microbial protein production as well as energy intake and partitioning by the animal (Allen, 2014). The type and temporal supply of fuels are affected by the type and source of carbohydrates, as well as interactions with other diet and animal factors. Physically effective NDF (peNDF) is the fraction of NDF in the diet with particles of a size adequate to form a rumen mat, which entraps small, potentially degradable particles in the rumen and enhances rumination. The increased re- tention time of digesta in the rumen increases total-tract digestibility and provides additional buffering capacity, likely reducing risk of low ruminal pH and a depression in NDF digestibility.

NEUTRAL DETERGENT FIBER

NDF is the most common measure of fiber used for rou- tine feed analysis. It is a simple and inexpensive method that measures most, but not all, of the chemical compounds that comprise the fiber fractions that are indigestible by mammalian enzymes. NDSF components such as pectins, P-glucans, fructans, and gums are not included in the NDF fraction. NDF has largely replaced other measures of fiber. Crude fiber does not quantitatively recover hemicellulose and lignin, and acid detergent fiber (ADF) does not include hemicellulose but includes some soluble fiber (e.g., pectin) unless ADF is done sequentially on NDF residue (Van Soest, 1994). Within a specific feedstuff, concentrations of NDF and ADF are highly correlated, but the correlation is lower for mixed diets that contain different fiber sources.

The concentration of NDF in feeds or diets is correlated negatively with energy concentration because NDF is generally less degradable than the NDS fractions of feeds. However, digestibility of NDF is highly variable depending on source, ruminal retention time, and the ruminal environ- ment. The acid detergent lignin (ADL) fraction and some of the hemicellulose and cellulose associated with lignin are essentially indigestible by bacterial and mammalian enzymes (Van Soest, 1994). Accordingly, NDF digestibility is negatively associated with lignin concentration within feed type, although the relationship is not consistent. Lignin does not affect the digestibility of the NDS fraction of feeds. Therefore, it is most useful to express ADL on an NDF basis. The ADL content of forages increases with maturity and is

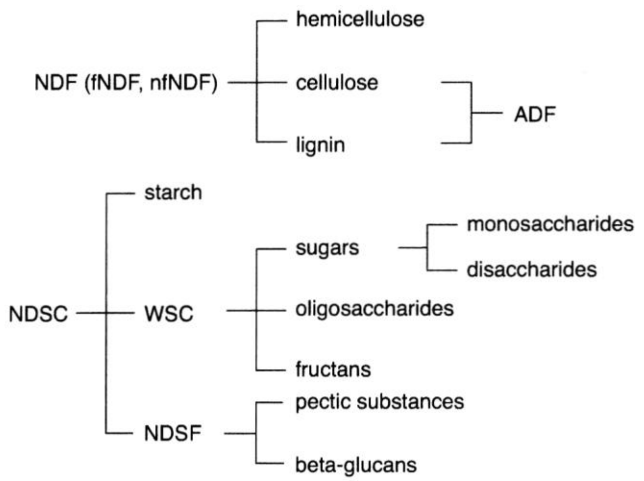

FIGURE 5-1 Carbohydrate fractions in feeds include NDF and NDSC. ADF is a component of NDF.

greater for legumes than grasses at similar stages of maturity (Smith et al., 1972), and warm-season grasses generally have greater ADL concentration than cool-season grasses (Man- debvu et al., 1999).

NEUTRAL DETERGENT-SOLUBLE CARBOHYDRATES

NDSC are diverse and include water-soluble carbohy- drates, starch, and NDSF. Except for the monosaccharides, lactose, malto-oligosaccharides, glycogen, and starch, all other NDSC must be degraded to monosaccharides or fer- mented by gut microbes to be of nutritional value to the animal. The concentration of NDSC in feeds has been estimated by difference as nonfiber carbohydrates (NFC), which is calculated as 100 percent of dry matter (DM) minus the sum of NDF, crude protein (CP), crude fat (CF), and ash; sometimes the CP and ash in NDF are subtracted from NDF to prevent redundant correction for ash and CP (NRC, 2001). Although NFC has been used as an estimate of NDSC, its use is not recommended for diet formulation. It aggregates analytical errors of each of the component assays. Furthermore, it incorrectly conveys the idea that a single pool of NDSC is uniform in its digestion and fer- mentation characteristics.

WSC include mono-, di-, oligosaccharides, and some poly- saccharides. Sugars include monosaccharides (glucose, fructose, etc.) and disaccharides (sucrose, lactose). Lactose is found specifically in milk products, whereas other sugars are found in many feeds, including cane and beet molasses (50 to 60 percent of DM), fresh forages (2 to 10 percent of DM), vegetable pulps (12 to 40 percent of DM), candy or bakery by-products (variable), and other processed human food by-products. Oligosaccharides such as stachyose (2 to 7 percent of DM) and raffinose (1 to 2 percent of DM) are found in soybeans (Kumar et al., 2010). Cool-season grasses are the primary source of fructans (0 to 30 percent of DM), a storage carbohydrate consisting of varying chain lengths of fructose sometimes with a single glucose at the reducing end of the chain.

Starch contains polymers (amylose and amylopectin) of glucose units linked by bonds that can be cleaved by mam- malian enzymes. It comprises the majority of the NDSC in feeds derived from grain crops and tubers that are generally increased in the diet to meet the energy demands of lactat- ing dairy cows. The primary sources of starch fed to cows are corn, barley, wheat, oats, sorghum, millet, silages made from the associated plants, and tubers such as cull potatoes. Concentrations of starch and NDF in corn and sorghum si- lages are inversely related because starch production during kernel filling dilutes the fibrous stover fraction of the plant (see feed tables, Chapter 19). Forages are supplemented with cereal grains to increase energy density, provide glucose precursors, and decrease the filling effects of rations. The starch concentration of lactating cow diets ranges from less than 20 percent to more than 30 percent of DM.

The most common NDSF components encountered in ru- minant feeds are pectins and mixed-linkage P-glucans. Pec- tins are composed of a backbone molecule made primarily of galacturonic acid and varying amounts of side chains made of arabinose, galactose, and other sugars. Primary sources of pectin in dairy cattle diets include legume forages, soy- bean hulls, and citrus and sugar beet pulps. Mixed-linkage P-glucans have the same structure as cellulose, except for a periodic bend in the linear glucose chain caused by a β-(1,3) linkage. These P-glucans are found in some small grains such as barley. In some feeds such as citrus pulp, NDSF concentrations can reach 40 percent of DM. Currently, NDSF is calculated by subtracting the concentrations of NDF and starch from the concentration of 80 percent ethanol insoluble residue where NDF and ethanol residues are express on an ash- and CP-free basis (Hall et al., 1999). Using 80 percent ethanol rather than water as a solvent removes some shorter- chain fructans from NDSF.

RUMINAL AND TOTAL-TRACT DIGESTION

Soluble fiber and WSC have the potential to be nearly completely degraded in the rumen, whereas ruminal digest- ibilities of NDF and starch are lower and highly variable by source and processing. WSC are very rapidly utilized by mixed ruminal microbes, and few of these carbohydrates likely escape ruminal utilization. Reported in vivo rates of ruminal disappearance of glucose, sucrose, and lactose were greater than 250 percent/h, and no residual glucose, fructose, or sucrose was detected in duodenal digesta 1 hour postdos- ing into the rumen (Weisbjerg et al., 1998). Although sugars are rapidly fermented in the rumen, there is little evidence that moderate substitution of sugars for starch in diets de- creases ruminal pH. Inclusion of sugars in diets by replacing starch generally either increases ruminal pH or has no effect for several possible reasons, including less acid produced per mole of hexose fermented and storage of glucose as microbial glycogen (Oba, 2011). The WSC may be partially converted to microbial glycogen that can be fermented in the rumen or pass to the small intestine. Rate of disappear- ance of grass fructans and raffinose is also rapid in mixed batch culture (Thomas, 1960), and orchardgrass fructan dis- appeared at 62 percent/h in vitro (Hall and Weimer, 2016). The disappearance rates of NDSF from pectin-rich citrus, sugar beet pulp, and alfalfa ranged from 20 to 40 percent/h in vitro (Hall et al., 1998), and pectin isolated from alfalfa showed similar rates of fermentation in vitro (Hatfield and Weimer, 1995). Unlike WSC, the digestibility of NDSF is reduced at low pH (Strobel and Russell, 1986). The avail- ability and ruminal digestion of WSC and NDSF are not as markedly affected by processing as is starch.

Ruminal digestibilities of starch and NDF are typically lower than soluble fiber and WSC and are highly variable. Ruminal fermentability of starch from various cereal grains ranges from less than 30 percent to more than 90 percent (Nocek and Tamminga, 1991; Firkins et al., 2001). It is af- fected by many factors, including grain type, endosperm vit- reousness, processing (e.g., rolling, grinding, steam-flaking), conservation method (e.g., dry, ensiled), ration composition, and animal characteristics. Wheat, barley, and oats have starch that is more readily degraded than corn starch, and sorghum starch is most resistant to degradation in the ru- men and digestion by the animal (Huntington, 1997). Based on a meta-analysis (Ferraretto et al., 2013), ruminal starch digestibility was greater for wheat (79 percent) and barley (71 percent) than for corn (54 percent), whereas total-tract starch digestibility did not differ (93 to 94 percent). Better descriptions of grain particle size and other discriminating variables will help characterize ruminal digestibility differ- ences within categories. The basis for these differences is related to the characteristics of protein in the endosperm of each grain. Starch granules in the endosperm of seeds are embedded in a protein matrix, and endosperm proteins vary in solubility and resistance to digestion (Kotarski et al., 1992). Vitreous endosperm in some grains contains prolamin proteins (e.g., zein in corn and kafirin in sorghum) that are insoluble and resistant to digestion, decreasing enzyme ac- cess to starch granules; in contrast, the proteins in floury endosperm are readily solubilized, allowing greater access to starch (Hoffman and Shaver, 2011). The vitreousness of corn grain endosperm is a function of genetics and maturity. Corn grain vitreousness increases with maturity (Phillipeau and Michalet-Doreau, 1997) and ranges from 0 percent to more than 75 percent at full maturity (Hoffman et al., 2010). Because corn silage and high-moisture corn are harvested before physiological maturity, their degree of vitreousness is less than that of dry shelled corn. When grains are ensiled, ruminal fermentability of starch can be affected by both grain moisture concentration (af- fected by maturity at harvest) and storage time. In vitro starch digestibility (IVSD) of high-moisture corn samples increased by 9 percentage units from October to August of the follow- ing year (Ferraretto et al., 2014). Furthermore, IVSD was negatively related to DM concentration and positively related to concentration of ammonia-N, likely because endosperm proteins are solubilized over time, increasing starch ferment- ability. The increase in protein solubility and IVSD over time is greatest for grains with higher moisture concentration, and changes were greatest over the first months of storage (Allen et al., 2003). The likely increase in ruminal starch digestibility of ensiled feeds over time should be considered when formulating diets.

To a greater degree than has been noted for other NDSC, the rate of starch hydrolysis is increased by more extensive processing, with greater response for grains with more vit- reous endosperm, such as sorghum and corn (Huntington, 1997). Processing increases access of enzymes to starch granules by reducing particle size, which increases surface area, or by swelling and disrupting kernel texture by steam- flaking. Reducing particle size by cracking and grinding significantly increases rate of starch degradation (Galyean et al., 1981; McAllister et al., 1993; Ferraretto et al., 2013), and the effect is greater with unprocessed than with heat- processed grains. Ruminal starch digestibility of ensiled (64 percent) and steam-flaked or steam-rolled (59 percent) corn was numerically greater than dry corn (54 percent) in a meta-analysis (Ferraretto et al., 2013). However, the number of treatment means for ensiled and steam-flaked/rolled corn was small compared with dry shelled corn, and treatment effects were not significant. In that study, total-tract starch digestibility was positively related to ruminal starch digest- ibility across starch sources. Total-tract starch digestibility was also positively related to postruminal starch digestibility (percentage of duodenal flow), but the wide range in total- tract starch digestibility from less than 84 percent to greater than 98 percent indicated that digestion postruminally does not completely compensate for starch that escapes ruminal degradation. Ruminal and total-tract starch digestibilities were linearly related with each unit increase in total-tract starch digestibility, corresponding to a 3.4 percentage unit increase in ruminal starch digestibility.

Ruminal digestibility of starch is also affected by the starch concentration of diets. In lactating cows, the fractional rate of starch digestion as well as ruminal digestibility of starch increased when corn grain was substituted for fNDF (Oba and Allen, 2003a) or nonforage fiber sources (NFFS; beet pulp; Voelker and Allen, 2003). Ruminal amylolytic enzyme activity for low-starch (21 percent) diets was calculated as 68 percent of that for high-starch (32 percent) diets (Oba and Allen, 2003a). The increases in starch digestibility and calcu- lation of higher amylolytic enzyme activity with higher levels of starch feeding could be caused by an increased number of amylolytic bacteria in the rumen (Mackie and Gilchrist, 1979). Furthermore, 6-hour starch digestibility in vitro of various feeds averaged 32 percent when using inoculum from cows fed a 50:50 concentrate/hay diet but only 7 percent with inoculum from a strictly hay-fed cow (Cone et al., 1989). Greater starch digestibility with higher-starch diets indicates that starch degradability in the rumen does not follow first- order kinetics and is a function of both the source as well as characteristics of the microbial population in the rumen.

NDF is degraded primarily in the reticulorumen, with additional degradation occurring in the large intestine. Across studies reported in the literature, ruminal digestibility of NDF accounts for over 90 percent of total-tract NDF digestibility (Huhtanen et ah, 2010; Gressley et ah, 2011). However, significant compensatory digestion can occur postruminally when ruminal NDF degradation is suppressed (Oba and Al- len, 2000b). Increasing starch in the diet typically depresses ruminal (Firkins et ah, 2001 ; White et ah, 2016) and total-tract (Ferraretto et ah, 2013) NDF digestibility. The digestibility of NDF depends on characteristics intrinsic to the NDF source that affect the maximal rate and extent of digestion, retention time in the fermentation compartments, and concentrations and activity of microbial enzymes (Allen and Mertens, 1988). Ruminal retention time is a function of characteristics of the animal (e.g., dry matter intake [DMI]) and feed (e.g., particle size, fragility, digestion characteristics).

The chemical composition of NDF varies greatly, which affects its susceptibility to enzymatic degradation. Ligni- fication of NDF varies among forages and NFFS and is negatively related to digestibility (Van Soest, 1994). The in- digestible NDF (iNDF) fraction is positively related to lignin concentration, and potentially digestible NDF (pdNDF) is the fraction remaining (NDF - iNDF); both can be expressed on a DM or NDF basis. The rate of pdNDF degradation in vitro is adversely affected by low pH and inclusion of starch (Grant and Mertens, 1992). Oba and Allen (2003b) reported a positive linear relationship between mean ruminal pH and degradation rate of pdNDF; degradation rate declined from

~4 percent/h at pH 6.5 to -1 percent/h at pH 5.7.

The actual digestibility of pdNDF is positively related to rate of degradation and the length of time microbial enzymes have to cleave the bonds. The NDF in plant tissues (e.g., mesophyll, xylem, phloem) degrades at different rates (Akin, 1989; Wilson, 1991), and their maximal rate of NDF degradation depends on chemical composition and anatomi- cal structure that affects accessibility of substrates to enzymes (Jung and Allen, 1995). Thus, surface area likely limits rate of NDF degradation unless other factors such as ruminal pH inhibit fibrolytic activity (Russell et al., 2009). Perennial C3 grasses generally have greater NDF concentration than legumes, as well as lower concentrations of lignin and iNDF. Although the rate of digestion of pdNDF is generally slower for grasses compared with legumes (Smith et al., 1972), ruminal digestibility is usually greater because of the greater pdNDF concentration and longer retention time in the rumen (Voelker Linton and Allen, 2008; Kammes and Allen, 2012a). Ruminal digestibility of NFFS is highly variable, depending on composition, rumen pH, and the peNDF of the diet, which affects their retention in the rumen (Firkins, 1997). Ruminal degradation in vivo proceeds at less than the maximal rate because concentrations and activity of enzymes are limited by environmental effects (pH, nutrient availability) on microbial populations. Availability of rumen-degradable protein (RDP) can limit NDF digestibility in continuous culture (Griswold et al., 2003), and the limitation of diet RDP on ruminal NDF digestibility appears to vary among feeds (Soliva et ah, 2015). Decreasing RDP decreased DMI quadratically (especially be- low 9.2 percent of DM) and tended to decrease ruminal ADF digestibility linearly (Reynal and Broderick, 2005). The direct effect of RDP on ruminal NDF digestibility has not been well studied in lactating dairy cows. Decreasing metabolizable protein (both RDP and rumen-undegradable protein [RUP]) significantly decreased total-tract NDF digestibility (Lee et ah, 2011, 2012), most likely as a result of decreased RDP limiting ruminal NDF digestibility.

Greater DMI is associated with decreased rumen retention time and digestibility of NDF (Riewe and Lippke, 1970); total-tract NDF digestibility decreased 4.4 percentage units as DMI increased from 2.5 to 5.0 percent of body weight in a meta-analysis with lactating Holstein cows (de Souza et al., 2018). Oba and Allen (1999a) reported a negative linear relationship between responses in DMI and total-tract NDF digestibility among 32 cows when they were fed diets con- taining brown midrib corn silage (bm3) or its near-isogenic control com silage.

Digestibility of dietary NDF can be depressed when ruminal retention time is reduced such as when high proportions of finely chopped or pelleted forages (Allen, 1997) or when NFFSs (Grant, 1997) are included in diets. Forage fragility varies greatly and affects the rate of reduction in particle size from chewing during eating and ruminating (Poppi et ah, 1981). Faster particle size reduction will increase the mass of particles below the threshold size to pass from the reticulorumen and decrease the ability of the rumen to selec- tively retain those particles by decreasing the mass of large fibrous particles in the rumen (Kammes and Allen, 2012d). Diets with more digestible fiber within a forage type (e.g., brown midrib corn silage and less mature grasses and alfalfa) tend to have shorter ruminal retention times (Oba and Allen, 2000a; Kammes and Allen, 2012b; Kammes et ah, 2012a) and grasses are retained longer than alfalfa (Voelker Linton and Allen, 2008; Kammes and Allen, 2012a, d). Length of cut did not affect the pool size or retention time of NDF in the rumen for orchardgrass (Kammes et ah, 2012b) or alfalfa- based diets (Kammes and Allen, 2012c). This might be because peNDF entering the rumen after chewing was either similar or above a critical threshold for ruminal retention of particulate matter between diets.

Absorbed nutrients produced from ruminal degradation and fermentation or intestinal digestion of carbohydrates in the gastrointestinal tract have different effects on DMI and energy partitioning. The primary nutrients that ruminal degradation provides to the animal are volatile fatty acids (VFAs) and microbial protein. The VFA can provide up to 70 percent of the energy required by cattle (Bergman, 1990), with the types and amounts produced affecting how the animal's glucogenic or ketogenic needs are met. Ruminal concentrations of the VFA do not indicate amounts of production because information on rumen liquid volume, passage, and differences in rates of absorption for the VFA is lacking (Sutton et al., 2003). Increasing the starch concentration of diets at the expense of NDF can greatly increase propionic acid production without affecting the production of acetic or butyric acids (Sutton et al., 2003), but starch passing from the rumen may be digested to glucose that is absorbed or metabolized to lactate in the small intestine (Reynolds et al., 2003). Ruminal fermentation of sugars generally increases the production of butyric acid (Oba, 2011), but ruminal fer- mentation of pectin yields mainly acetic acid and relatively little propionic acid (Dehority, 1969). Although the VFA produced can differ by carbohydrate type (Murphy et al., 1982; Weimer, 2011), they are also affected by pH (Strobel and Russell, 1986), dietary forage inclusion (Murphy et al., 1982), concentration of RDP (Malestein et al., 1984), amount of organic matter (OM) degraded, and likely other undefined factors. Propionate production from starch has been reported to differ between diets with higher or lower amounts of for- age (Murphy et al., 1982). As dietary starch is increased and pH declines, populations of amylolytic/lactate-producing and lactate-utilizing bacteria both increase (Mackie and Gilchrist, 1979). As starch supplementation increases, the lactate utilizers likely convert a greater proportion of lactate to propionate (Baldwin et al., 1962; Aschenbachet al., 2011).

PREDICTION OF RUMINAL CARBOHYDRATE DIGESTIBILITY

Ruminal fermentation of carbohydrates affects produc- tivity and is a critical consideration for diet formulation. Whereas efforts to predict ruminal carbohydrate digest- ibility have been abundant, there is considerable uncertainty remaining. Chemical analyses (e.g., NDF, starch) have improved, but a better understanding and description of carbohydrate degradation and digestion will likely improve future predictability. Ruminal digestibility values used in diet formulation are usually derived from feed dictionaries or values reported in the literature, single time-point incuba- tions, rumen models based on rates of digestion and passage, or empirical equations based on diet factors and DMI.

Feed Libraries

Ruminal digestibility of NDF and starch in vivo is highly variable. Whereas some of this variation can be accounted lor by feed type, and means can be included in tables of nu- trient composition, table values do not account for the large variation resulting from the effects of growing environment, genetics, conservation method, and processing on the feeds, limiting their usefulness, especially for feeds with highly vari- able composition such as forages and some by-product feeds.

Single Time-Point Incubations

Digestibility of NDF and starch in vitro or in situ by ruminal microbes is often measured at single time points. These methods can provide important relative information to compare feeds, but they are less useful to predict ruminal digestibility in vivo. Ruminal digestibility is affected by ruminal pH and enzyme activity, which vary with the diet and its interaction with animal factors, so rates of digestion are affected by method of determination (e.g., in vitro, in situ; Krizsan et al., 2012). Total-tract NDF digestibility was related to in vitro NDF digestibility with an incubation time of 48 hours ($r = 0.55$, $P = 0.01$) but not 30 hours in a small data set of 21 diets from seven experiments, but total-tract NDF digestibility was overestimated by ~7 percentage units, and the bias was greater as digestibility increased, indicating that equations will be needed to convert in vitro digestibil- ity into in vivo digestibility (Lopes et al., 2015). Although in vitro NDF digestibility at single time points may not be acceptable for prediction of in vivo digestibility, they have been positively related to response in DMI and milk yield (Oba and Allen, 1999b) and are best used to compare feeds for allocation to different groups of cows, troubleshooting, or purchasing considerations.

Mechanistic and Empirical Rumen Models

Mechanistic models have been developed and have evolved over the past several decades to predict ruminal carbohydrate digestibility to more accurately predict metabo- lizable energy and metabolizable protein in diet formulation programs. The basic concept of mechanistic rumen models is that digestion of OM in the rumen is a competition be- tween the rates of degradation and passage (Waldo et al., 1972). The major problem with this approach is the lack of accurate data for rates of degradation and passage of indi- vidual feeds or nutrient fractions in vivo. Whereas feeds can be fractionated and rates of degradation of fractions can be measured, the rates obtained do not represent actual rates in vivo because of differences in particle size (surface area), enzyme activity, and pH between measurement conditions and in the rumen of cows (Firkins et al., 1998; Krizsan et al., 2012). Also, accurate passage rates for each nutrient fraction within specific feeds that correspond to its rate of digestion are nonexistent. The use of the same overall passage rate for all fractions within feeds will overestimate ruminal digest- ibility of soluble fractions and small particles that have faster rates of passage and will underestimate ruminal digestibility of large particles that have much slower rates of passage.

Models of digestion in, and passage from, the rumen that have been developed over the past several decades such as Molly (Baldwin et al., 1987; Hanigan et al., 2013) and the Cornell Net Carbohydrate and Protein System (Russell et al., 1992; Sniffen et al., 1992; Van Amburgh et al., 2015) and its derivatives and have made valuable contributions. They have helped students understand the complex interac- tions of the diet, microbes, and the animal; codify research to understand rumen function; stimulate and prioritize ideas for new research; and stimulate ideas to solve prob- lems during diet formulation. However, whereas they have these advantages over empirical models, they are also less accurate because of incomplete knowledge, numerous re- quired inputs, and lack of accurate data for parameterization (France et al., 2000).

Prediction of carbohydrate digestion with empirical equa- tions gives up some of the potential advantages of mechanis- tic models based on rates of digestion and passage in favor of increased accuracy. Empirical equations have been devel- oped to predict ruminally degraded starch and NDF based on diet composition and DMI (White et al., 2016) and are used to predict microbial protein production (see Chapter 6).

PHYSICALLY EFFECTIVE NEUTRAL DETERGENT FIBER

A sufficient mass of long, fibrous particles is required to optimize digestive efficiency and proper rumen function. The particles form a mat in the rumen that acts as a filtration bed to increase retention of potentially escapable particles containing pdNDF, increasing NDF digestibility as well as digesta mass in the rumen to buffer fermentation acids. A greater mass of digesta increases rumen buffering both directly, by cation ex- change, and indirectly, by stimulating rumen movements that enhance mixing and absorption of fermentation acids, as well as by stimulation of rumination and salivary buffer secretion (Allen, 1997). Other benefits include reduced risk of abomasal displacement and a greater ability to maintain euglycemia and reduce mobilization of body reserves when feed intake decreases in the short term (Allen and Piantoni, 2014).

peNDF is that fraction of the diet that contributes to the for- mation of this retention mechanism. Whereas increased peNDF concentration of the diet can increase buffering and NDF digestibility, the increased digesta mass can also reduce DMI when limited by ruminal distension. Effectiveness of NDF can be determined multiple ways, but the most accepted definition is the ability of feeds to stimulate chewing (Sudweeks et al., 1981; Allen, 1997). Researchers have assessed whether milk fat can be used as an index of “effective” NDF (e.g., Grant, 1997; Caccamo et al., 2014), which includes the “physically effec- tive” (pe) component that stimulates chewing plus the chemical component of NDF that reduces ruminal acid production when substituted for starch (Armentano and Pereira, 1997). However, because other factors such as bioactive fatty acids (Lock et al., 2006) and feeding of sugars (Oba, 2011) that are not associated with peNDF affect milk fat, use of milk fat production to assess sufficiency of peNDF should be done with caution.

Within the NDF fraction, fNDF that has not been finely pro- cessed is the primary contributor to peNDF of diets, and rumen pH is positively related to fNDF but not total NDF concentra- tion of diets (Allen, 1997). Forages that are long or coarsely chopped provide NDF in a form that is more physically effec- tive than NDF in NFFS such as soyhulls or com gluten feed, or NDF in cereal grains. Many NFFS have a large fraction of potentially digestible NDF, small particle size, and relatively high specific gravity (Batajoo and Shaver, 1994). Furthermore, they have similar or faster passage rates than forages (Bhatti and Firkins, 1995) with similar or slower degradation rates (Firkins, 1997). Based on chewing, NFFS NDF was 30 to 80 percent as effective as fNDF across studies (Mertens, 1997). However, peNDF values of feeds are determined relative to the forage used in each experiment and can vary from one experi- ment to another. For instance, physical effectiveness of whole cottonseed was 50 percent compared with long-theoretical length of cut (9.5 mm) but 127 percent compared with short (4.8 mm) alfalfa silage (Mooney and Allen, 1997).

Evaluation of treatment means from experiments that compared particle length within the same forage indicated a breakpoint for effect of forage particle length on total chew- ing time per day at a mean sieve aperture size of 0.3 cm (Al- len, 1997). Therefore, a threshold above which little additional response in chewing time to increased peNDF concentration of the diet is likely, and further increases might limit DMI (Zebeli et al., 2012). Using peNDF of individual feeds in ra- tion formulation followed by evaluation of the peNDF of the total mixed ration (TMR) is useful because of the potential for particle size reduction during mixing, especially with ver- tical mixers. Mertens (1997) suggested that peNDF of rations be measured by determining the proportion of TMR particles retained on a 1.18-mm screen and multiplying by the total NDF concentration of the diet. However, this assumes that diet DM and NDF are equally distributed across all screens if done on an as-fed basis, which is rarely the case. Another limitation to using particle size measurements is the lack of standard methodology used across experiments with use of both wet- and dry-sieving with different sieve designs. The Penn State Particle Separator (PSPS) was introduced as an inexpensive and rapid method to characterize particle size (Lammers et al., 1996) and is a common method of determin- ing particle size of forages and TMRs on farms.

A meta-analysis approach detected associations between peNDF retained on and above a screen with an 8-mm ap- erture (peNDF >8) and rumen pH, time < pH 5.8 (h/d), and rumination time (min/d) with a breakpoint at 18.5 percent peNDF >8 (Zebeli et al., 2012). That analysis further dem- onstrated that the requirement for peNDF must be balanced with effects of peNDF on DMI because DMI was decreased when peNDF >8 exceeded 14.9 percent, suggesting diets should contain between 14.9 and 18.5 percent peNDF >8 depending on whether maximizing intake or rumen pH was the goal. Although the relationships were reasonably strong, the database used for the meta-analysis did not include many diets with NFFS, which have been used to lessen both fNDF and starch concentrations while maintaining DMI and milk production (Dann et ah, 2015). When peNDF is calculated by multiplying the fraction of the diet over a threshold particle size by the NDF of the entire diet, the resulting value will in- crease by adding any source of NDF, even if its physical effec- tiveness is zero. This problem could be solved by calculation of peNDF as the NDF concentration of the fractions above a threshold size (e.g., what is retained on an 8-mm screen), as suggested by Zebeli et al. (2012). Data have not been ad- equately reported in the literature at this time to evaluate this, and it could not be rapidly and routinely measured on farms. The peNDF of the DM consumed is also affected by sorting, which increases with excessive large particles, particularly in dry rations (Leonardi and Armentano, 2003; Kmicikewycz et ah, 2015). For these reasons, peNDF should be considered as an optimal range to target rather than a minimum that can be exceeded without potential negative consequences.

RECOMMENDATIONS

The previous committee (NRC, 2001) concluded that the application of the effective fiber concept was limited because of a lack of standard, validated methods to measure effective fiber of feeds and to establish requirements. Progress has been made since that publication but is still limited by the availability of published research reporting the particle size distribution of dietary NDF needed for a robust implemen- tation. Because of this, the committee recommends using fNDF as one option for estimating physical form adequacy of diet in combination with other factors.

Forage Neutral Detergent Fiber

fNDF is recommended as a primary consideration for diet formulation rather than total NDF because of a greater positive relationship with ruminal pH (Allen, 1997) and a greater negative relationship with DMI (Allen, 2000). The minimum fNDF likely varies from 15 to 19 percent of diet DM and depends on the proportion of total NDF, starch, and perhaps other NDSC in the diet. The average effective value of NDF from nonforage sources was set to 50 percent of that for NDF from forage. For every 1 percentage unit decrease in NDF from forage (as a percentage of dietary DM) below
19 percent, the recommended concentration of total NDF was increased 2 percentage units, and maximum starch was reduced 2 percentage units (see Table 5-1). Data are needed to determine whether concentrations of WSC and NDSF affect NDF requirements. The minimum total NDF was set at 25 percent based on studies cited in the previous edition (NRC, 2001) and comes with caveats (i.e., the forage was assumed to have adequate particle size, dry ground corn was the predominant starch source, and cows were fed a TMR).

The optimal fNDF concentration of diets to maximize energy intake is higher than the minimum to reduce risk of acidosis. Optimal fNDF for lactating cows likely ranges from 17 to 27 percent of diet DM and is a function of milk yield and the cows' drive to eat as well as other factors shown in Figure 5-2. Less-filling diets will likely benefit cows with high milk yield with DMI limited by ruminal distention by allowing greater feed intake while maintaining rumen fill. However, the greater energy concentration of the diet might result in less rumen fill for cows with lower milk yield and DMI limited by metabolic mechanisms (Allen, 2000). The filling effect of fNDF is not constant but is affected by the initial size and fragil- ity of forage particles, which affect ruminal retention time and formation of the rumen mat (Allen, 2000). The optimal fNDF concentration of diets also depends on diet fermentability (Al- len, 1997), which is highly dependent on the concentration and fermentability of starch. At a given fNDF concentration, diet fermentability can be decreased by substituting grains (i.e., starch) with NFFS or by substituting sources of starch that are less fermentable such as dry ground com for high- moisture com, wheat, or barley. Diet fNDF concentration must increase when high percentages of highly fermentable starch

TABLE 5-1 Recommended Minimum Forage and Total NDF and Maximum Starch Concentration of Diets for Lactating Cows When a Diet Is Fed as a TMR, the Forage Has Adequate Particle Size, and Dry Ground Corn Is the Predominant Starch Source

Minimum fNDF	Minimum Total NDF	Maximum Starch
19	25	30
18	27	28
17	29	26
16	31	24
15	33	22

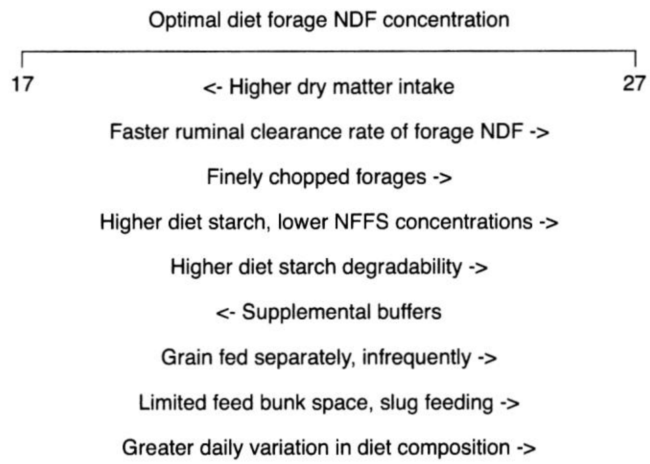

FIGURE 5-2 Factors affecting the optimal forage NDF concen- tration of diets for lactating cows. Clearance rate of forage NDF from the rumen is affected by rate of degradation, forage fragility, and rate of passage.
SOURCE: Adapted from Allen (1995).

sources (e.g., wheat, barley, high moisture corn) are included in diets, whereas supplemental buffers will allow lower fNDF concentration. The optimal fNDF is also affected by feeding method; TMR decreases variation in rumen pH over time compared with feeding grains separately (Robinson, 1989), allowing lower fNDF diets with higher starch concentration. For component feeding systems, when concentrate is offered more than twice daily, fewer effects on production, milk com- position, and ruminal conditions have been reported (Cassel et al., 1984; Robinson, 1989; Maltz et al., 1992). Eating rate is decreased with increased fNDF concentration (Oba and Al- len, 2000a) or forage particle size (Kammes et al., 2012b), and optimal fNDF will be greater when competition for feed bunk space encourages some cows to slug feed. fNDF concentration should be increased when greater variation in carbohydrate fractions is expected. Variation might be from differences in forage DM or NDF that are not adjusted for by diet formulation. Although intentional variation in fNDF concentration of diets did not affect production responses of early to mid-lactation cows, treatment diets averaged 23.2 percent fNDF and always exceeded 21 percent fNDF, and cows were not fed diets with lower fNDF diets for more than a couple of days (Yoder et al., 2013). Variation not accounted for by reformulation of diets will likely result in negative effects (e.g., milk fat depression, decreased NDF digestion) when diets are closer to the recom- mended minimum fNDF concentration.

Physically Adjusted Neutral Detergent Fiber

A potential option for assessing adequacy of the physical form of diets is the physically adjusted NDF (paNDF) system (White et al., 2017a,b), which was developed to provide guid- ance on diet particle size requirements to attain a given rumen pH and its interaction with other diet components, including diet NDF, fNDF, starch, and percent forage. However, this system is new and has not undergone rigorous field testing. The new system was termed "paNDF" to avoid confusion with peNDF, which is based on stimulation of total chewing. Particle size recommendations are based on the PSPS (Lam- mers et al., 1996) because of the available data and its wide use on farms. In the paNDF system, many variables describing particle size, composition of the diet, predictions of ruminally degraded carbohydrates, and more are inputs to predictions for DMI, rumination time, and ruminal pH. Although the ef- fect of high paNDF on limiting DMI is important, the focus of the system is only on the role of paNDF to maintain the mean ruminal pH selected. The ruminal pH goals within the paNDF system are proxies for describing a desirable rumen environ- ment and are not intended to be predictive of ruminal pH or recommendations for optimal ruminal pH. Ruminal pH reflects the net production, absorption, metabolism, and neutralization of fermentation acids plus other factors (Allen et al., 2006). Two mean ruminal pH targets (6.0 and 6.1) are represented in Table 5-2 to illustrate the interactions among the model inputs. At the database's average ADF/NDF (0.63), the discrete values of 3,9, and 15 percent of DM in the TMR on the 19-mm sieve were chosen to reflect the range in the model's predictiveness. At each of those discrete cutoffs, the minimum percentage of DM on the PSPS 8-mm screen to maintain the mean ruminal pH shown is predicted. Particle size characteristics can be interpolated in diets with varying model terms (forage, starch, NDF, and fNDF percentages). As in Table 5-1, there is a con- sistent relationship in which less fNDF is needed to maintain any categorical mean pH as starch declines.

Assuming limited sorting of feed by the animals, an ex- ample of how the paNDF system works would be if a new forage source is coarser, then the TMR would be expected to have higher percentage of DM on the 19- and 8-mm screens; in such a case, the percentage of fNDF could be decreased without an expected negative impact on ruminal pH. In con- trast, if a coarser forage is replaced by more finely ground forage, the fNDF inclusion in the diet should be increased. Similarly, a more conservative (i.e., higher) percentage of particles on the 19-mm sieve could be selected when ruminal starch digestibility is expected to be greater than average because of more extensive grain processing.

The impact of ADF/NDF is demonstrated in Figure 5-3, when other terms are held constant; as dietary fNDF decreases below 20 percent, a greater percentage of particles is needed on the 8-mm sieve with increasing ADF/NDF, which is partly affected by grass/legume (Allen and Piantoni, 2014) but also af- fected by other factors such as percent cereal grains in the diet.

Caveats for the Physically Adjusted Neutral Detergent Fiber System

The paNDF system was not designed to predict ruminal pH but, rather, to demonstrate how chemical composition and physical form of the diet affect rumen pH. In turn, improved understanding should ultimately guide ration formulation and TMR evaluation. Although a variety of pH targets can be chosen within this system, results appear reliable only between pH 6.0 and 6.1, which is the recommended range in usage. The range among individual studies in the derivation data set was 5.44 to 6.83 (White et al., 2017a), but the model included random effects that reduce variation among studies and is therefore sensitive to small changes in mean ruminal pH. In addition, derived solutions are sensitive to BW and CP because these factors affect DMI, which is used to predict total time ruminating. When evaluating paNDF it is recommended to assume a diet of 17 percent CP and BW of 630 kg so that changes in diet factors related to paNDF remain within condi- tions included in the data from which the models were derived.

The paNDF system was based on studies that primarily were from individually fed dairy cattle and therefore do not reflect the greater potential to sort against long or dry forage or toward grain (Kmicikewycz et al., 2015; Miller-Cushon and DeVries, 2017) compared with commercial settings or over- crowding conditions that limit feed bunk space and encourage slug feeding. Factors shown in Figure 5-2 must be considered

TABLE 5-2 Predicted Minimum Dietary DM on the 8-mm Sieve of the PSPS with Varying DM on the 19-mm Sieve Predicted to Maintain Mean Ruminal pH ≥6.0 or ≥6.1 for Lactating Dairy Cattle Fed TMR Varying in Dietary Composition[a]

	TMR composition, % (DM Basis)[b]			Maintenance of pH 6.0 % TMR on 19-mm Sieve			Maintenance of pH 6.1 % TMR on 19-mm Sieve		
				3	9	15	3	9	15
Forage	Starch	NDF	fNDF	Minimum % on 8-mm Sieve[c]			Minimum % on 8-mm Sieve"		
40	35	25	19	20	13	12	53	43	33
40	30	28	19	26	17	14	53	42	33
40	25	30	22				17	10	11
40	25	30	19	19	12	11	36	26	19
40	25	30	17	32	23	17	50	40	31
40	20	33	17	11	10	10	19	12	11
40	20	33	15	24	15	12	32	22	16
40	20	33	14	30	21	15	39	29	21
40	15	35	13				17	10	10
50	35	25	20	25	17	13	40	30	22
50	35	25	18	40	31	23	59	48	38
50	30	28	22	12	10	10	23	14	12
50	30	28	20	26	17	13	38	28	20
50	30	28	18	41	31	23	54	44	34
50	25	30	22				17	10	10
50	25	30	20				31	22	15
50	25	30	18	24	15	12	46	36	27
50	20	33	19				18	10	10
50	20	33	17				32	22	16
60	30	28	23				42	32	23
60	30	28	22				51	41	31
60	25	30	24				22	13	10
60	25	30	23				30	20	14
60	25	30	22				38	28	19
60	20	33	20				26	17	12

[a]Total mixed rations (TMRs) from the Penn State Particle Separator (PSPS) with only two sieves and a pan (White et al., 2017a,b). These two pH targets are provided as examples, and this approach is not intended to be used to predict ruminal pH.

[h] fNDF = forage NDF. Other variables in the model were set to their means (White et al., 2017a). The mean ADF/NDF was 0.63 (SD = 0.26) in the overall data set (used for this table) and 0.56 (0.07), 0.63 (0.14), and 0.63 (0.08) for 40, 50, and 60 percent forage, respectively.

[c] When a row is blank, pH 6.0 is predicted to be achieved at all possible combinations of particle size percentages; that is, there is no minimum on the 8-mm sieve to predict a mean ruminal pH of 6.0.

in addition to the paNDF system with the nutrition advisor's experience and judgment. Ruminal buffers and other minerals might influence mean ruminal pH independent of any of the diet variables shown, particularly for diets that have high amounts of rumen-degraded starch. For situations in which high-forage diets are fed (such as typical diets fed to dry cows), the paNDF system is not necessarily useful and should not be used to predict rumination time or DMI (outside of the data range). In the meantime, the committee recommends further research on PSPS fractions, including differences in moisture and NDF concentrations among diets and on other parameters that could improve the paNDF model or lead to the develop- ment of other particle size-based systems.

Feed Analysis

Forages should be tested routinely for concentrations of DM and NDF because of the large variation among and within sources. Starch concentration should also be tested routinely for com silage, small grain silages, and other variable starch sources. Frequent sampling and testing will help determine more accurate values because of other sources of variation, including sampling and laboratory errors. Frequency of test- ing depends on inclusion rate, the variation expected, and the length of time the forage will be fed. Abrupt changes in quality are more likely to occur in upright silos compared with horizontal silos, with small fields of forage, and when weather during harvest is less than optimal. Each lot of NFFS that is typically variable (e.g., distillers grains) should be tested before inclusion in rations, especially if inclusion rate is high unless feeding rates on the farm make this impractical.

Testing forages for in vitro NDF digestibility is recom- mended for allocating multiple forages to different groups of cows, troubleshooting, or purchasing considerations. In vitro digestibility is a biological measure and more variable across runs than chemical measures. Therefore, comparisons should be made within a run or results corrected to a standard that is replicated within runs. Because perennial grasses are

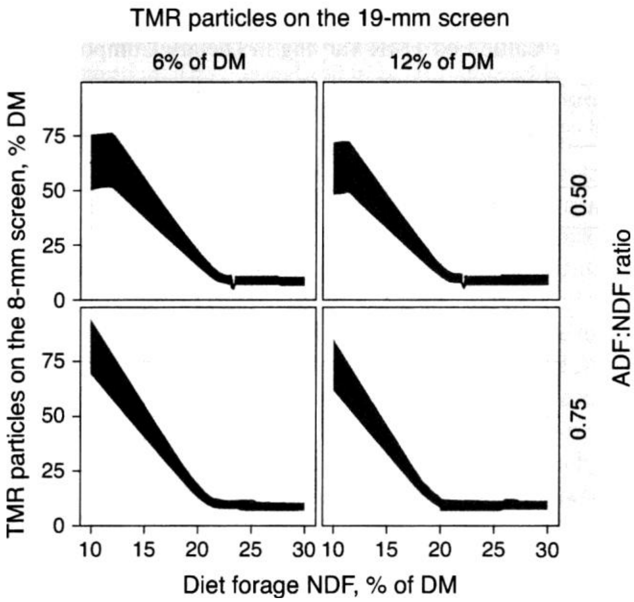

FIGURE 5-3 Example of how the dietary ratio of ADF/NDF (0.50 for the top two plots or 0.75 for the bottom two plots) influences the paNDF model of White et al. (2017b). A ratio of 0.5 is typical of a grass-based diet or a diet with substantial amounts of by-products, and a ratio of 0.75 is more typical of a diet based on corn silage and alfalfa. The line represents the ensemble prediction, and the shaded area represents the minimum to maximum range of predictions from individual ensemble members as an estimated confidence range. The mean ADF/NDF in that database was 0.63. When the total mixed ration (TMR) particles on the 19-mm sieve of the PSPS are fixed at 6 percent (left two plots) or 12 percent (right two plots) of DM, the TMR recovered on the 8-mm sieve (left axis, percentage of DM) is the minimum needed to achieve a mean ruminal pH of 6.1 with varying dietary forage NDF (percentage of DM; bottom axis). Other variables in the model are held constant to their means. SOURCE: White et al. (2017b).

generally more filling than legumes (Oba and Allen, 1999b), mixed grass-legume forages should be tested for ADF to help determine the fraction of grass and legume in the forage; ADF/NDF is ~0.8 for legumes and -0.6 for grasses (Allen and Piantoni, 2014).

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